ENERGY INFRASTRUCTURE ECONOMICS, POWER, FUELS AND ENERGY SECURITY

ENERGY-SERVICE VALUE RAIL → POWER-ENERGY DISTINCTION → ELECTRICITY-LAYER MAP → REAL-TIME BALANCE BOARD → INSTITUTIONAL PERIMETER → TARIFF-FINANCE SPLIT

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00

ENERGY-SERVICE VALUE RAIL

ENERGY-SERVICE VALUE RAIL | PRIMARY RESOURCE | CONVERSION + NETWORK | DISTRIBUTION + END USE | RELIABLE AFFORDABLE SERVICE | MUST REMEMBER | ENERGY INFRASTRUCTURE CONNECTS PRIMARY ENERGY, CONVERSION, GENERATION

CORE CLAIM
- PRIMARY RESOURCE / IMPORT
- CONVERSION + NETWORK

MECHANISM
- DISTRIBUTION + END USE
- RELIABLE AFFORDABLE SERVICE

CONSEQUENCE / LIMIT
- MUST REMEMBER: Energy infrastructure connects primary energy, conversion, generation,...
- PRIMARY RESOURCE / IMPORT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Energy infrastructure connects primary energy, conversion, generation,.

01

STAGE 01

POWER-ENERGY DISTINCTION

POWER-ENERGY DISTINCTION | POWER CAPACITY | ENERGY FLOW | CAPACITY SHARE | GENERATION SHARE | TOTAL ENERGY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MW / GW → power capacity	CAPACITY SHARE != generation share	MW / GW → power capacity
MWh / GWh → energy flow	ELECTRICITY != total energy	MWh / GWh → energy flow

CORE CLAIM
- MW / GW → power capacity
- MWh / GWh → energy flow

MECHANISM
- CAPACITY SHARE != generation share
- ELECTRICITY != total energy

CONSEQUENCE / LIMIT
- MW / GW → power capacity
- MWh / GWh → energy flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAPACITY SHARE != generation share; ELECTRICITY != total energy.

02

STAGE 02

ELECTRICITY-LAYER MAP

ELECTRICITY-LAYER MAP | TRANSMISSION + SYSTEM OPERATION | DISTRIBUTION + RETAIL | MARKET + REGULATION

TRANSMISSION + SYSTEM OPERATION

DISTRIBUTION + RETAIL → MARKET + REGULATION

CORE CLAIM
- TRANSMISSION + SYSTEM OPERATION
- DISTRIBUTION + RETAIL

MECHANISM
- MARKET + REGULATION
- TRANSMISSION + SYSTEM OPERATION

CONSEQUENCE / LIMIT
- DISTRIBUTION + RETAIL
- MARKET + REGULATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSMISSION + SYSTEM OPERATION; DISTRIBUTION + RETAIL; MARKET + REGULATION.

03

STAGE 03

REAL-TIME BALANCE BOARD

REAL-TIME BALANCE BOARD | GENERATION + IMPORT + DISCHARGE | DEMAND + EXPORT + CHARGE + LOSS | RESERVE + FLEXIBILITY PROTECT BALANCE

CORE CLAIM
- GENERATION + IMPORT + DISCHARGE
- DEMAND + EXPORT + CHARGE + LOSS

MECHANISM
- RESERVE + FLEXIBILITY protect balance
- GENERATION + IMPORT + DISCHARGE

CONSEQUENCE / LIMIT
- DEMAND + EXPORT + CHARGE + LOSS
- RESERVE + FLEXIBILITY protect balance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEMAND + EXPORT + CHARGE + LOSS; RESERVE + FLEXIBILITY protect balance.

04

STAGE 04

INSTITUTIONAL PERIMETER

INSTITUTIONAL PERIMETER | TECHNICAL PLANNING | RETAIL SERVICE | CEA → TECHNICAL PLANNING

CEA → technical planning

CERC / SERC → regulation → GRID-INDIA / SLDC → operation → DISCOM → retail service

CORE CLAIM
- CEA → technical planning
- CERC / SERC → regulation

MECHANISM
- GRID-INDIA / SLDC → operation
- DISCOM → retail service

CONSEQUENCE / LIMIT
- CEA → technical planning
- CERC / SERC → regulation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID-INDIA / SLDC → operation; DISCOM → retail service.

05

STAGE 05

TARIFF-FINANCE SPLIT

TARIFF-FINANCE SPLIT | EXPLICIT STATE SUBSIDY | REGULATORY ASSET | DISCOM DEBT | CLOSE DISTINCTION | CAPACITY IN MW IS NOT GENERATION IN MWH, PLANT

EXPLICIT STATE SUBSIDY

REGULATORY ASSET → DISCOM DEBT / UNDER-RECOVERY → CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not

CORE CLAIM
- EXPLICIT STATE SUBSIDY
- REGULATORY ASSET

MECHANISM
- DISCOM DEBT / UNDER-RECOVERY
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not...

CONSEQUENCE / LIMIT
- EXPLICIT STATE SUBSIDY
- REGULATORY ASSET

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not.

06

STAGE 06

DISCOM DIAGNOSTIC

DISCOM DIAGNOSTIC | ENERGY RECOVERY | REVENUE-COST GAP | PPA + SUBSIDY + COLLECTION | ONE RATIO CANNOT DIAGNOSE ALL STRESS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
AT&C → energy recovery	PPA + SUBSIDY + COLLECTION	AT&C → energy recovery
ACS-ARR → revenue-cost gap	ONE RATIO cannot diagnose all stress	ACS-ARR → revenue-cost gap

CORE CLAIM
- AT&C → energy recovery
- ACS-ARR → revenue-cost gap

MECHANISM
- PPA + SUBSIDY + COLLECTION
- ONE RATIO cannot diagnose all stress

CONSEQUENCE / LIMIT
- AT&C → energy recovery
- ACS-ARR → revenue-cost gap

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PPA + SUBSIDY + COLLECTION; ONE RATIO cannot diagnose all stress.

07

STAGE 07

PETROLEUM PERIMETER

PETROLEUM PERIMETER | PIPELINE + STORAGE | REFINING + MARKETING | PNGRB EXCLUDES CRUDE + GAS PRODUCTION

CORE CLAIM
- UPSTREAM → production
- MIDSTREAM → pipeline + storage

MECHANISM
- DOWNSTREAM → refining + marketing
- PNGRB excludes crude + gas production

CONSEQUENCE / LIMIT
- UPSTREAM → production
- MIDSTREAM → pipeline + storage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DOWNSTREAM → refining + marketing; PNGRB excludes crude + gas production.

08

STAGE 08

ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON

ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON | PHYSICAL AVAILABILITY | INFRASTRUCTURE ACCESSIBILITY | USER AFFORDABILITY | SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

PHYSICAL AVAILABILITY

INFRASTRUCTURE ACCESSIBILITY → USER AFFORDABILITY → SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

CORE CLAIM
- PHYSICAL AVAILABILITY
- INFRASTRUCTURE ACCESSIBILITY

MECHANISM
- USER AFFORDABILITY
- SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

CONSEQUENCE / LIMIT
- PHYSICAL AVAILABILITY
- INFRASTRUCTURE ACCESSIBILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFRASTRUCTURE ACCESSIBILITY; USER AFFORDABILITY; SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE.

09

STAGE 09

RENEWABLE-INTEGRATION CHAIN

RENEWABLE-INTEGRATION CHAIN | VARIABLE CAPACITY | GRID + FORECAST | FLEXIBILITY + STORAGE | RELIABLE GENERATION VALUE

VARIABLE CAPACITY

GRID + FORECAST → FLEXIBILITY + STORAGE → RELIABLE GENERATION VALUE

CORE CLAIM
- VARIABLE CAPACITY
- GRID + FORECAST

MECHANISM
- FLEXIBILITY + STORAGE
- RELIABLE GENERATION VALUE

CONSEQUENCE / LIMIT
- VARIABLE CAPACITY
- GRID + FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID + FORECAST; FLEXIBILITY + STORAGE; RELIABLE GENERATION VALUE.

10

STAGE 10

STATUS AND UNIT LADDER

STATUS AND UNIT LADDER | INSTALLED CAPACITY | GENERATION + SERVICE OUTCOME | ANNOUNCED / BILL

ANNOUNCED / BILL

APPROVED / SANCTIONED / OUTLAY → INSTALLED CAPACITY → GENERATION + SERVICE OUTCOME

CORE CLAIM
- ANNOUNCED / BILL
- APPROVED / SANCTIONED / OUTLAY

MECHANISM
- INSTALLED CAPACITY
- GENERATION + SERVICE OUTCOME

CONSEQUENCE / LIMIT
- ANNOUNCED / BILL
- APPROVED / SANCTIONED / OUTLAY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: APPROVED / SANCTIONED / OUTLAY; INSTALLED CAPACITY; GENERATION + SERVICE OUTCOME.

11

STAGE 11

ENERGY-ECONOMICS ANSWER SPINE

ENERGY-ECONOMICS ANSWER SPINE | DEFINE CARRIER + UNIT + LAYER | TRACE NETWORK + TARIFF + REGULATOR | TEST SECURITY + TRANSITION | CONCLUDE SERVICE, NOT HEADLINE CAPACITY | EVIDENCE LIMIT | STATE FUEL/TECHNOLOGY, STOCK OR FLOW, UNIT

DEFINE carrier + unit + layer → TRACE network + tariff + regulator → TEST security + transition → CONCLUDE service, not headline capacity → EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit

CORE CLAIM
- DEFINE carrier + unit + layer
- TRACE network + tariff + regulator

MECHANISM
- TEST security + transition
- CONCLUDE service, not headline capacity

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,...
- DEFINE carrier + unit + layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | DISTRIBUTIONAL AND TRANSITION COST | OVERCONSUMPTION IF VARIABLE PRICE IS TOO LOW | EXCLUSION IF FIXED CHARGE IS TOO HIGH | REVENUE INSTABILITY IF FIXED NETWORK COST IS RECOVERED MAINLY | SHIFTS BURDEN TO FUTURE CONSUMERS | CAPACITY NO LONGER NEEDED | OLD HIGH TARIFFS AFTER TECHNOLOGY COST FALLS

OPTIONAL SOURCE DEPTH
- distributional and transition cost
- overconsumption if variable price is too low;
- exclusion if fixed charge is too high;
- revenue instability if fixed network cost is recovered mainly per kWh while rooftop/self

ADVANCED DEBATE
- shifts burden to future consumers;
- capacity no longer needed;
- old high tariffs after technology cost falls;
- fuel/transport assumptions that later fail.

LEGACY / NUANCE
- performance and availability incentives;
- transmission readiness before commissioning.
- suppress wholesale energy prices;
- reduce thermal running hours;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels